

Link to repository: [Ojas2011/Telco-Customer-Churn-Prediction](https://github.com/Ojas2011/Telco-Customer-Churn-Prediction)

Introduction:

In a company called Telco, there are many customers who have different kinds of feedback to give on different things. And through that feedback we can see what kinds of things the organization is lacking and what kinds of things the organization needs to work on, but what if we could predict based on the feedback whether a customer would return or not given their response to different questions? One such survey taken from at least seven thousand people was put into a dataset in a github repository. Therefore, my research question was that: based on customer reviews and their responses to different questions about Telco, what is the likelihood of them returning as customers again? Through this we will answer and predict how customer feedback is connected to customer churn.

Data:

The data used in this project was found from the github repository containing the dataset 'Telco-Customer-Churn'. It has 7043 rows which show the 7043 different customers and their different opinions/satisfaction with the services, hence showing a diverse dataset with customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Churn.

Below is a formal description of the variables(taken from the github repository):

- customerID: ID of specific customers
- gender: Gender of customer
- SeniorCitizen: Whether the customer is a senior citizen or not
- Partner: Whether the customer has a partner or not
- Dependents: Whether the customer is a dependent or not
- tenure: Tenure of a customer
- PhoneService: Which phone service customer uses(if they use it)
- MultipleLines: Whether customer has multiple lines
- InternetService: Which internet service(if used) is used by a customer
- OnlineSecurity: Whether customer has online security
- OnlineBackup: Whether customer has online backup
- DeviceProtection: Whether customer has device protection
- TechSupport: Whether customer has tech support
- StreamingTV: Whether customer has/uses streaming TV
- StreamingMovies: Whether customer has/uses streaming movies
- Contract: Description of customer contract
- PaperlessBilling: Whether customer has paperless billing
- PaymentMethod: Payment method used by the customer
- MonthlyCharges: Monthly charges for a customer

- TotalCharges: Total amount of charges for a customer
- Churn: Whether customers want to return or not

The dependent variable of this dataset is churn, which has two unique variables “No” for no returning of customer and “Yes” for returning of customer.

Preprocessing and Exploratory Data Analysis:

After reviewing the initial dataset and coding it on Visual Studio Code, there were no null values present. Because there is a huge split on customers saying “Yes” and customers saying “No” (People saying yes in 1000s and people saying no in 5000s) there had to be some cleaning done to keep it unbiased.

Initial Exploratory Data analysis showed an imbalance. The dataset contains 5174 customers saying they wouldn't like to continue with Telco and 1869 customers saying they would like to continue. This shows that the statistics may not fully be unbiased, however later on we did get back a decision tree accuracy of 77.11 percent.

There are also other biases with a significant split between other variables such as online security where 3498 have no online security, 2019 do and 1526 have no internet in the first place which would cause bias since online security is a variable to be taken into account, but if someone does not have internet in the first place what answer can they give, there are many such biases in this dataset.

Modeling:

To ensure we minimize the bias and maximize the accuracy, we cleaned the variables/responses in places where they were not necessary such as people having no internet services were out of the dataset for things such online security and tech support variables since those not having internet in the first place can't really help with questions about tech support. We also had to remove the customer ID everywhere since it had nothing to do with customer churn or any other variable. While tenure was a minor variable and therefore the data for it was not cleaned it is still an important factor since the amount of years a specific person was with a company impacted their trust in it and their willingness to stay with it regardless of latest updates.

Out of the many models used, some important ones were RandomForestClassifier, DecisionTreeClassifier, LabelEncoder, classification_report. Performance was evaluated using the feature importances used in Random Forest, the accuracy data contained in the Logistic Regression and the precision, recall, f1-score, support, churn/no churn, accuracy, macro average, weighted average in the decision tree accuracy.

Results:

Of the models mentioned, Random Forest had an accuracy of 0.7861, Decision Tree had an accuracy of 0.7164, Logistic Regression had an accuracy of 0.7939 which means that Logistic Regression performed better than the other two. The reason logistic regression did better most likely has to do with fact that type of learning it used with the dataset. This shows that the model did a relatively decent job of seeing how different variables impacted customer churn for Telco.

When seeing what impacts customer churn, monthlyCharges and totalCharges play a good role, including but not limited to the contract and tenure in which higher charges will impact whether the customer wants to return or not as well as how long their tenure and contract is.

Resources:

Nas-virat. (n.d.). *Telco-customer-churn* [Dataset]. GitHub.

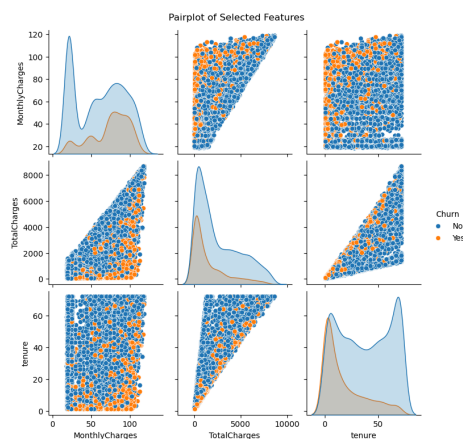
<https://github.com/Nas-virat/Telco-Customer-Churn>

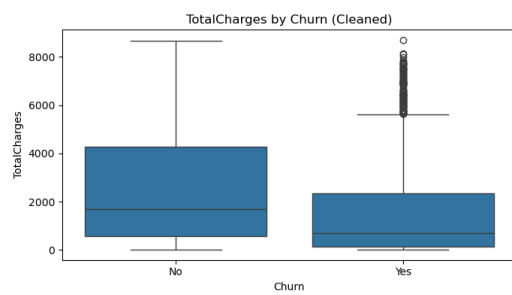
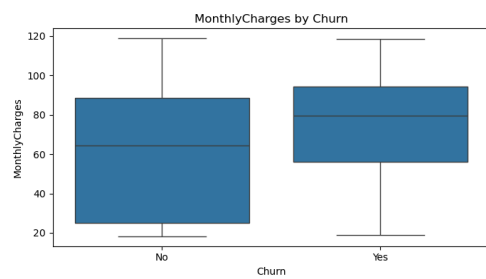
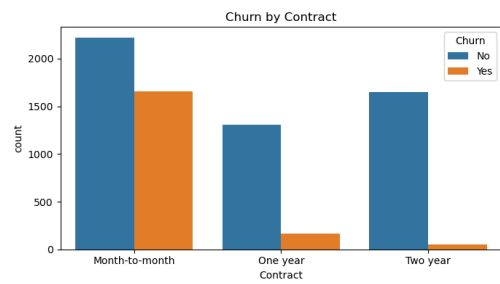
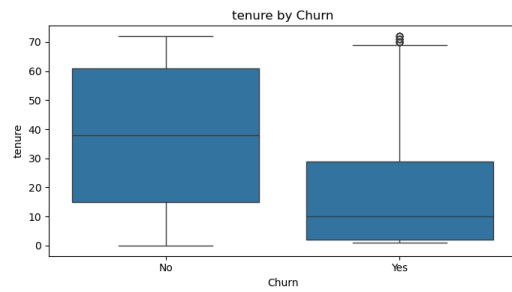
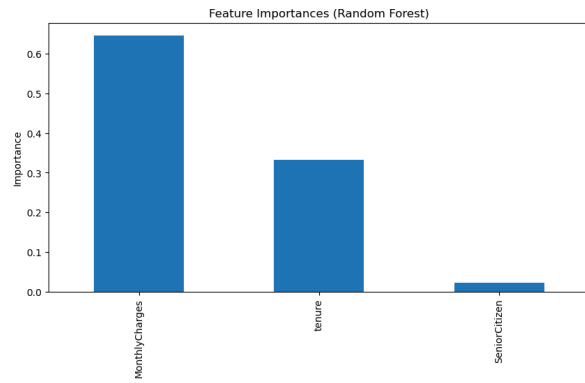
- I also used the scikit learn part 2 Jupyter Notebook code for reference on some parts of the code.

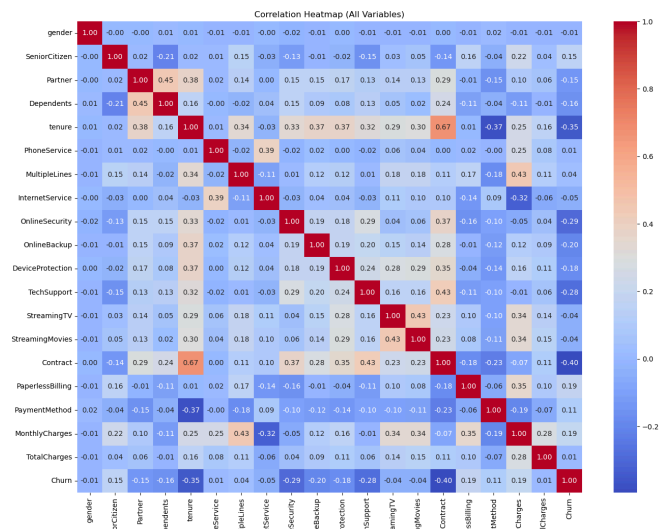
AI Acknowledgement:

- Github Copilot was used to bring all my ideas for visualizations and statistics to write the code for both.

Appendix:







	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

Decision Tree Accuracy: 0.7711				
	precision	recall	f1-score	support
No Churn	0.81	0.90	0.85	1033
Churn	0.60	0.41	0.49	374
accuracy			0.77	1407
macro avg	0.71	0.66	0.67	1407
weighted avg	0.75	0.77	0.76	1407